2-1-1: Examining Network Cabling Issues

At the end of this episode, I will be able to:

1. Identify and explain common network cabling issues, given a scenario.

Learner Objective: *Identify and explain common network cabling issues, given a scenario*

Description: In this episode, the learner will examine common cabling issues facing today's networks. We will explore interference and mismatch issues and copper and fiber optic-based cabling issues.

- *We have seen that interference can hinder Wi-Fi signals, can this happen in cabling?*
    - External signals can disrupt a cable's signal.
    - Electromagnetic sources like motors, fluorescent lights, HVAC and elevators.
    - Signal degradation
        - a reduction in the quality of the signal as it travels through the cable.
    - Potential solutions
        - Use shielded cables
        - Reroute cables away from interference sources
- *The motors, fluorescent lights and few more, how do these interfere?*
    - This increases the potential for electromagnetic interference or EMI, or unwanted noise/electrical energy in the twisted pair cabling.
    - Can be an incorrect cabling choice due to environmental awareness
        - Shielded Twisted Pair (STP) vs. Unshielded Twisted Pair (UTP)
        - STP provides additional electromagnetic interference protection.
        - Use STP in high interference environments; UTP in cost-sensitive, lower interference scenarios.
- *You mentioned external signals causing interference, how does this happen?*
    - Crosstalk
        - is when the signal from one cable leaks into another.
    - Common causes
        - Physical cable damage
        - Tightly packed cables
        - Lack of or poor shielding
        - Poor quality cables
    - Potiential solutions
        - Regular cable inspections
        - use high-quality cables
        - Cable management techniques
- *What about mismatches?*
    - Can be an incorrect cable
        - The wrong type of cable for the network's requirements/hardware capabilities
        - Such as a standards mismatch such (CAT 5e vs. 6 vs. 6a vs. 8)
            - Categories differ in data rate and bandwidth support.
            - Misuse arises from outdated standards knowledge.
        - Lack of understanding of cable specifications or network requirements
    - Can be TX/RX mismatches
        - Mixing up the transmit (TX) and receive (RX) ends of a cable.
        - Cable mismatch (straight through vs. crossover)
        - Improper termination, TIA 568 mismatch
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        - Common cause - misunderstanding or lack of cable wiring standards
    - Potential solutions
        - Educate and train staff on cable types and network requirements
        - Select category based on network speed and future-proofing considerations.
        - Ensure cables are clearly labeled
        - Provide clear up-to-date documentation
- How about fiber optic cabling?
    - Common fiber optics issues
        - Physical damage
            - Bending or pulling can cause breaks or microbeads in the fibers
        - Connector issues
            - Improperly cleaned or aligned connectors
            - Can cause significant loss and reflection problems
        - Environmental factors
            - Temperature fluctuations and moisture
            - Can affect fiber performance and longevity
    - Single-Mode Fiber (SMF) Issues
        - Installation Sensitivity
            - Requires precise alignment and handling due to its narrow core diameter
            - Installation and maintenance can be more challenging
        - Higher equipment costs
            - Compatible equipment (like lasers)
            - Costs more than those used with MMF.
        - Attenuation and dispersion
            - While less than MMF, still can be affected by bending and physical stress.
    - Multimode Fiber (MMF) Issues
        - Modal dispersion
            - Leads to signal distortion over long distances
            - This can limit bandwidth and data transmission rates.
        - Distance limitations
            - Effective only for short-range communications
            - Limited use in extended networks
        - Core diameter variability
            - Differences in core sizes can lead to mismatches and loss when interconnecting different types or generations of MMF

- Additional Resources:
    - Dispersion
        - Refers to the broadening of light pulses as they travel through the fiber
        - This affects signal quality and transmission rates.
    - Modal Dispersion
        - Occurs in multimode fiber optics when different light modes travel at different speeds,
        - This leads to signal distortion and reduced bandwidth over longer distances.